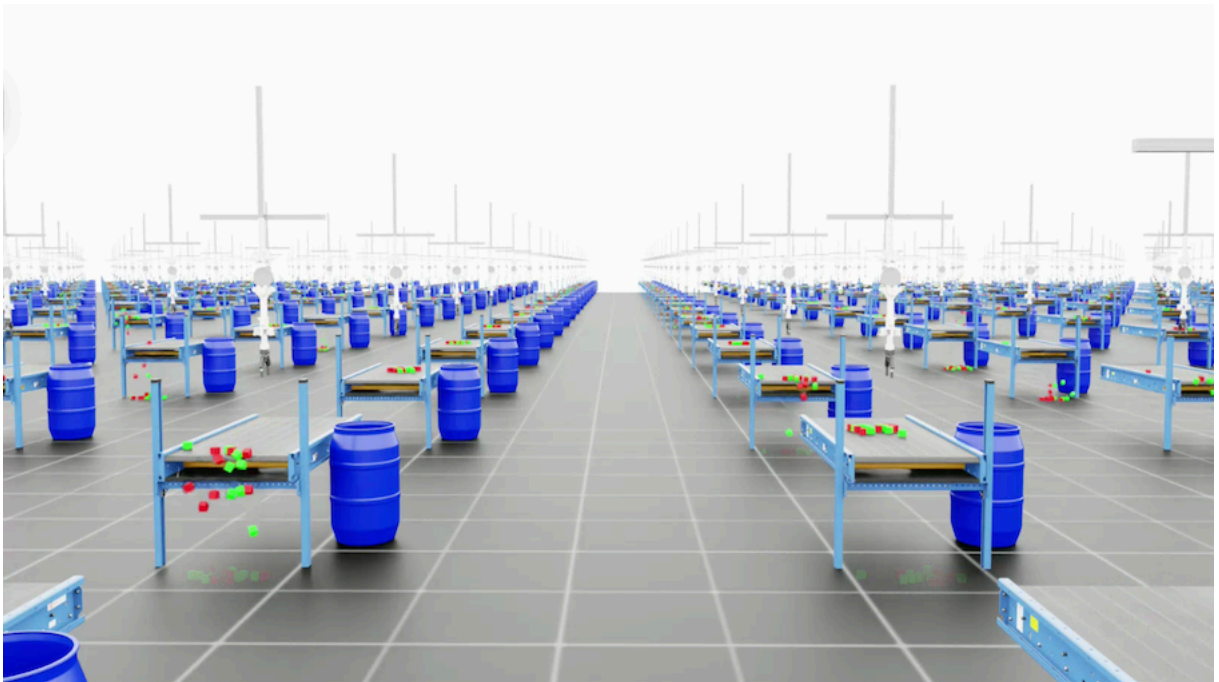


Simulative Construction and Verification of a Tendon-Driven Robot for Conveyor-Based Pick-and-Place Tasks Using Isaac Lab

Project Thesis in the program Computational Engineering M. Sc.

Friedrich-Alexander-Universität Erlangen-Nürnberg
Lehrstuhl für Fertigungsautomatisierung und Produktionssystematik
Prof. Dr.-Ing. J. Franke



Author: Georg Meyer

Student ID: 22791103

Supervisors:

Deadline: 01.07.2026

Duration: x months

Erklärung

Ich versichere, dass ich die vorliegende Arbeit ohne fremde Hilfe und ohne Benutzung anderer als der angegebenen Quellen angefertigt habe und dass die Arbeit in gleicher oder ähnlicher Form noch keiner anderen Prüfungsbehörde vorgelegen hat und von dieser als Teil einer Prüfungsleistung angenommen wurde. Alle Ausführungen, die wörtlich oder sinngemäß übernommen wurden, sind als solche gekennzeichnet.

Erlangen, den 04.04.2026

Georg Meyer

Table of Contents

Erklärung	3
Guideline	1
0.1 Format and Style	1
0.1.1 Page Layout and Typography	1
0.1.2 Page Numbering and Binding	1
0.1.3 Section and Chapter Formatting	1
0.1.4 Paragraphs, Spacing, and Whitespace	1
0.1.5 Figures	2
0.1.6 Tables	2
0.1.7 Diagrams	2
0.1.8 Formulas	2
0.1.9 Appendix	3
0.1.10 Eidesstattliche Erklärung (Statutory Declaration)	3
0.2 Thesis Structure and Content	3
0.2.1 Required Document Order	3
0.2.2 Page Count Guidelines	3
0.2.3 Red Thread (Roter Faden)	3
0.2.4 Chapter: Einleitung und Zielstellung (Introduction)	4
0.2.5 Chapter: Grundlagen und Stand der Technik (Background and State of the Art)	4
0.2.6 Chapter: Handlungsbedarf (Research Gap)	4
0.2.7 Chapter: Methodenteil (Methodology)	4
0.2.8 Chapter: Ergebnisteil (Results)	5
0.2.9 Chapter: Diskussionsteil (Discussion)	5
0.2.10 Chapter: Zusammenfassung und Ausblick (Summary and Outlook)	5
0.3 Scientific Writing Style	5
0.3.1 General Principles	5
0.3.2 Sentence Construction	5
0.3.3 Grammar	6
0.3.4 Technical Language	6
0.3.5 Expression and Phrasing	6
0.3.6 Content Guidelines	6
0.3.7 Introductions and Transitions Between Sections	6
0.3.8 Consistency and Formatting Rules	7
0.4 Citations and Bibliography	7
0.4.1 Citation Style	7
0.4.2 Citation Placement	7
0.4.3 Direct Quotes	7
1 Introduction	8
1.1 Motivation and Context	8

1.2 Problem Statement and Objectives	8
1.3 Thesis Outline	8
2 Background	9
2.1 Reinforcement Learning for Robotic Manipulation (2.1)	9
2.2 Physics Simulation in Isaac Sim 5.1 (2.2)	9
2.3 NVIDIA Isaac Sim and IsaacLab Architecture (2.3)	9
2.4 Tendon-Driven Robot Kinematics and Modeling (2.4)	9
3 State of the Art and Research Gap	10
3.1 GPU-Accelerated RL for Manipulation (3.1)	10
3.2 Tendon- and Cable-Driven Robot Simulation (3.2)	10
3.3 Workspace Analysis Methods (3.3)	10
3.4 Identified Research Gap (3.4)	10
3.5 Success Criteria (3.5)	10
4 Methodology	11
4.1 Overview of Approach (4.1)	11
4.2 Physical Robot Design (4.2)	11
4.2.1 Kinematic Chain and Degrees of Freedom (4.2.1)	11
4.2.2 Antiparallelogram Elbow Mechanism (4.2.2)	11
4.2.3 Cable-Driven Wrist (4.2.3)	11
4.2.4 Drive System and Tendon Routing (4.2.4)	11
4.3 Simulation Model Construction (4.3)	11
4.3.1 Mesh Processing Pipeline (4.3.1)	11
4.3.2 Articulation and Physics Configuration (4.3.2)	11
4.3.3 Disc Approximation Variant (4.3.3)	11
4.3.4 Physical Antiparallelogram Variant (4.3.4)	11
4.3.5 Five-DoF Assembly (4.3.5)	11
4.4 Tendon Actuation in Simulation (4.4)	11
4.4.1 Jacobian-Transpose Torque Mapping (4.4.1)	11
4.4.2 Physical Body-Force Tendon Model (4.4.2)	11
4.4.3 Actuation Mode Comparison (4.4.3)	11
4.4.4 Motor Specifications and Simulation Parameters (4.4.4)	11
4.5 Model Validation (4.5)	11
4.5.1 Step-Response Testing Protocol (4.5.1)	11
4.5.2 Monte Carlo Workspace Analysis (4.5.2)	11
4.6 Reinforcement Learning Task Formulation (4.6)	11
4.6.1 Simulation Environment (4.6.1)	11
4.6.2 Reach Task (4.6.2)	12
4.6.3 Cube Place Task (4.6.3)	12
4.6.4 Cube Sort Task — Outlook (4.6.4)	12
4.7 Training Infrastructure (4.7)	12
4.7.1 PPO Configuration (4.7.1)	12

4.7.2	Simulation Parameters (4.7.2)	12
4.7.3	Software and Hardware (4.7.3)	12
5	Results	13
5.1	Model Validation Results (5.1)	13
5.2	Reach Task Results (5.2)	13
5.3	Cube Place Results (5.3)	13
5.4	Cube Sort Preliminary (5.4)	13
5.5	Summary of Key Results (5.5)	13
6	Discussion	14
6.1	Evaluation of Outcomes (6.1)	14
6.2	Comparison with Related Work (6.2)	14
6.3	Limitations of the Study (6.3)	14
6.4	Practical Implications (6.4)	14
6.5	Recommendations and Lessons Learned (6.5)	14
7	Conclusion	15
7.1	Conclusion (7.1)	15
7.2	Outlook (7.2)	15
8	Appendices	16
	Bibliography	17

Guideline

0.1 Format and Style

0.1.1 Page Layout and Typography

- Paper format: white DIN A4 paper
- Font: Arial, 12pt (in this Typst template, Helvetica achieves this)
- Line spacing: 1.2 (the template uses 1.25)
- Text alignment: justified (Blocksatz) with automatic hyphenation enabled
- Font emphasis: use only in exceptional cases; headings follow the template style and must not be underlined
- Bullet points: only square bullet points (FAPS green), never round ones; use the template's built-in list style
- FAPS color scheme: FAPS-Grün (RGB 151, 193, 57) and white

0.1.2 Page Numbering and Binding

- Page numbering starts with the introduction (Arabic numerals)
- Directories (table of contents, list of figures, etc.) receive separate Roman numeral numbering; the cover page is not counted
- Print single-sided unless otherwise agreed with the supervisor
- Two bound copies required: adhesive binding with clear cover foil and green (FAPS green) back sheet
- Each copy must include a CD/DVD containing:
 - Digital version of the thesis (PDF)
 - All digital sources (papers, scanned documents, websites)
 - All images, tables, measurement data, models, and other data

0.1.3 Section and Chapter Formatting

- Chapters are numbered numerically with at most three levels (e.g., 1.1.1); no trailing dot after the last digit
- If a subsection X.1 exists, there must also be at least an X.2 (applies to all levels)
- Main chapters should have at least 3 subchapters; aim for a balanced structure (e.g., 5 chapters at level 1, each with ~4 at level 2, each with ~3 at level 3)
- Headings in the table of contents must match the headings in the text verbatim
- Every heading must be followed by actual content (including introductory text at the start of main chapters)

0.1.4 Paragraphs, Spacing, and Whitespace

- Each paragraph must consist of at least two sentences
- No blank line between paragraphs within a chapter; one blank line before the next heading at the end of a chapter
- No double spaces anywhere in the document; use search-and-replace to eliminate them before submission
- Use non-breaking spaces () between:
 - Titles and names: "Prof. Dr. Beispielhausen"
 - Date expressions: "19. September 2007"
 - Abbreviations: "z. B.", "u. a.", "d. h."

- Numbers and units: “44 mm”
- Numbers and words: “1.350 Personen”, “Nº 5”
- Citation references
- Check automatic hyphenation carefully for errors

0.1.5 Figures

- Every figure receives a caption below it (Unterschrift), formulated so the figure is understandable without the surrounding text
- All figures are numbered consecutively with Arabic numerals and listed in the list of figures
- Every figure must be referenced in the body text
- Title image: no caption, no numbering, not included in the list of figures
- Appendix figures: numbered within each appendix separately; not included in the list of figures
- Ensure sufficient image quality; recreate figures if originals are blurry, low-resolution, or scanned
- Use the same font in figures as in body text (Arial); font size 10pt is acceptable but must be consistent across all figures
- All figure components must be fully labeled: photo components, axes, parameters, legends with complete process parameters
- Fill areas with color only if the color conveys meaningful information; otherwise keep figures clean and minimal
- For recreated/modified figures from sources, cite with “in Anlehnung an [xy]” (based on [xy])
- Use a consistent color scheme across all figures; preferably FAPS colors unless industry cooperation requires otherwise
- Use a self-created graphic on the first page of the thesis

0.1.6 Tables

- Every table receives a caption above it (Überschrift) with a meaningful, descriptive title
- Tables are numbered consecutively with Arabic numerals and listed in the table directory
- Every table must be referenced in the body text
- Use the same font as in body text; maintain consistent font size across all tables
- Preferred alignment inside tables: left-aligned or centered (not justified)
- If a table spans a page break, do not break within a row; repeat the table header on the new page
- Extensive tables belong in the appendix

0.1.7 Diagrams

- Follow DIN 461 for axis labeling in diagrams
- Label axes with an arrow in the direction of increase (except for trial numbers etc.)
- Replace the axis label at the second-to-last tick mark with the unit
- Display all relevant information within the diagram
- Place a frame around diagrams
- Use Arial font inside diagrams with a legible font size
- Diagrams are captioned and numbered as figures (no separate numbering type)

0.1.8 Formulas

- Formula numbers are enclosed in round parentheses
- Numbering is chapter-based in the form X.Y (X = chapter number, Y = formula number within the chapter)
- Numbering restarts within each main chapter

- Variables used in a formula must be explained directly below the formula in tabular form

0.1.9 Appendix

- Technical drawings, extensive tables, supplementary documents, etc. go into appendices
- Each appendix starts on a new page and may span multiple pages
- Appendices are numbered with consecutive uppercase Latin letters (Appendix A, Appendix B, etc.)

0.1.10 Eidesstattliche Erklärung (Statutory Declaration)

- A signed declaration of independent authorship must be included on a separate page without a page number, placed after the cover page
- It must be signed by hand

0.2 Thesis Structure and Content

0.2.1 Required Document Order

1. Deckblatt (Cover Page)
2. Erklärung des Kandidaten (Statutory Declaration)
3. Inhaltsverzeichnis (Table of Contents)
4. Abbildungsverzeichnis (List of Figures)
5. Tabellenverzeichnis (List of Tables)
6. Abkürzungsverzeichnis (List of Abbreviations)
7. Einleitung und Zielstellung (Introduction and Objective)
8. Grundlagen / Stand der Technik (Background / State of the Art)
9. Handlungsbedarf (Research Gap / Need for Action)
10. Methodenteil (Methodology)
11. Ergebnisteil (Results)
12. Diskussionsteil (Discussion)
13. Zusammenfassung und Ausblick (Summary and Outlook)
14. Literaturverzeichnis (Bibliography)
15. Anhang (Appendix)
16. Lebenslauf (Curriculum Vitae)

0.2.2 Page Count Guidelines

- Bachelorarbeit: ca. 60–80 pages
- Projekt-/Studienarbeit: ca. 60–80 pages
- Master-/Diplomarbeit: ca. 80–120 pages
- These counts exclude appendices and directories

0.2.3 Red Thread (Roter Faden)

- A clear red thread must run through the entire thesis, starting from the central research question in the introduction
- The reader must always be able to see how any part of the thesis relates to the research question
- The thesis must proceed purposefully and not lose itself in tangential topics
- All claims and statements must be causally justified; results must not merely be listed but convincingly explained
- Each main chapter must begin with a short introduction motivating the content and end with a transition to the next

- Verify the red thread by checking all headings alone (without reading the text)—the logical flow should be recognizable

0.2.4 Chapter: Einleitung und Zielstellung (Introduction)

- Length: approximately 2 pages, maximum 3 pages
- Purpose: establish contact with the reader; provide orientation (“What can the reader expect?”)
- Content requirements:
 - Introduce the topic and its general background; explain why it is relevant
 - Present the specific research question
 - Justify the scope and boundaries of the thesis (what is included, what is excluded, and why)
 - Optionally describe the state of research (to show what exists and where gaps are)
 - Explain the methodological approach
 - Outline the structure of the thesis for the reader

0.2.5 Chapter: Grundlagen und Stand der Technik (Background and State of the Art)

- Clarify central concepts and define relevant terms
- Present the current state of the art objectively and without judgment (evaluation happens in the next chapter)
- Only cover theories, definitions, and knowledge directly relevant to the research question; avoid overly broad coverage
- If multiple competing theories exist, provide an overview
- Definitions are only needed for terms not commonly known by domain experts
- Present content in your own words; near-verbatim copying or translation from sources counts as insufficient
- Use primarily English-language sources where possible
- Prefer journal articles and conference papers over websites
- Avoid popular-science sources; only quote newspapers for current events
- Internet sources are acceptable only if no current literature exists elsewhere (e.g., software documentation); verify their credibility

0.2.6 Chapter: Handlungsbedarf (Research Gap)

- Critically discuss the literature from the previous chapter regarding unsolved problems and errors
- Derive current deficits and research gaps; formulate the specific research question
- Explain who considers the problem significant and what its broader impact is
- Describe the contribution the thesis makes to both research (theory, models, methods, facts) and practice
- Narrow the problem statement into a concrete, achievable objective with clear expected outcomes

0.2.7 Chapter: Methodenteil (Methodology)

- Together with Results and Discussion, this forms the main body (~2/3 of the thesis)
- Describe the investigation plan, procedures, and all steps and tools in sufficient detail for reproducibility
- Justify why specific methods were chosen over alternatives (pros/cons analysis)

- For experimental studies: describe the population and sample in all relevant characteristics; justify sample adequacy
- Scientific criteria for methods: objectivity, reliability, and validity
- The description must enable other researchers to reproduce the investigation
- Only describe analysis procedures in detail if they are novel or non-standard

0.2.8 Chapter: Ergebnisteil (Results)

- Present results of the investigation objectively and without interpretation
- Structure the chapter logically, aligned with the theory and methodology sections
- Use figures, tables, and diagrams to visualize results
- No interpretation or evaluation at this stage

0.2.9 Chapter: Diskussionsteil (Discussion)

- Summarize and critically discuss the results
- Evaluate and interpret results in relation to the original problem and objective from the introduction
- Argue the significance for scientific discourse and practical implications

0.2.10 Chapter: Zusammenfassung und Ausblick (Summary and Outlook)

- Maximum 3 pages
- Reconnect to the central problem and objective from the introduction
- Compactly summarize the most important results
- Place results in a broader context
- Address:
 - What are the key findings with respect to the research questions?
 - How can the results be interpreted or evaluated?
 - What questions remain open, and why?
 - What do the results mean for future work on this topic?
 - Where are opportunities for further research?
- Discuss the limitations of the investigation
- Provide suggestions for future work and improvements

0.3 Scientific Writing Style

0.3.1 General Principles

- Write in present tense (Präsens) and nominal style (Nominalstil)
- Use impersonal form throughout; never use “ich” (I) or “man” (one/you)
- Develop the red thread of argumentation before formulating sentences
- Every sentence and every claim must be precise and well-placed
- Write result-oriented: “Die Einflüsse führen zu Veränderungen.” not “Im Rahmen der Arbeit wurden untersucht...”

0.3.2 Sentence Construction

- Use passive voice sparingly—prefer active formulations
- Avoid sentences with “man” entirely
- Be careful with infinitive constructions
- Avoid chaining too many nouns together (“Substantivaneinanderreihungsproblem”)
- No nested or overly complex sentences (Schachtelsätze)

0.3.3 Grammar

- Avoid relative time references (“kürzlich”, “später”, “modern”) since the thesis should remain valid for years
- Do not use subjunctive mood (könnte, sollte, müsste, dürfte)

0.3.4 Technical Language

- Use precise adjectives; avoid vague terms like “innovativ” or “intelligent”
- No pleonasms (redundant word pairs)
- Do not artificially inflate terms
- Be cautious with superlatives and comparative forms
- Do not invent words
- Use anglicisms and foreign words sparingly; only when no equivalent German term exists
- Use precise technical terminology; avoid colloquial language
- Use technical terms consistently throughout

0.3.5 Expression and Phrasing

- Spell out numbers from zero to twelve; use digits from 13 onward
- Avoid colloquialisms and journalistic clichés
- Do not use rhetorical questions; make clear statements instead
- Avoid vague qualifiers: “im Allgemeinen”, “in der Regel”, “beispielsweise”, “teilweise”
- Avoid filler words: “nämlich”, “also”, “so”, “eben”
- Avoid repetition of both words and arguments
- Always formulate concretely and precisely; define unambiguously; enumerate completely; avoid generalizations
- Use nominal style

0.3.6 Content Guidelines

- Do not document iterative development processes or suboptimal intermediate solutions; present only the final approach
- Do not describe device/hardware development in excessive detail; focus on scientifically relevant, generalizable, transferable aspects
- Use direct quotes rarely and only pointedly; prefer evaluating and developing ideas further in your own words
- Introduce abbreviations only when necessary for space; spell out on first use and include in the abbreviation index
- Emphasize scientific quality rather than engineering challenges
- Emphasize fundamental, generalizable, and transferable findings rather than application-specific details
- Instead of giving examples, enumerate completely, structure, and evaluate
- Make all enumerations and representations as complete and non-overlapping as possible (MECE principle)
- Always support claims with your own analyses or cited sources
- No implied personal opinion and no subjective assessments except in the Outlook section
- Where possible, summarize each chapter’s key message in a figure for quick reference

0.3.7 Introductions and Transitions Between Sections

- Every main chapter or main section must begin with a brief introduction and end with an outlook/transition to the next section
- When introducing detailed explanations, name all relevant points in the introductory sentence

0.3.8 Consistency and Formatting Rules

- Introduce each abbreviation exactly once and then use it consistently
- Never write a paragraph consisting of only one sentence (minimum two sentences)
- Never write a list with only one bullet point (minimum two)
- Use nominal style for bullet-point items
- Maintain consistent spelling of expressions throughout

0.4 Citations and Bibliography

0.4.1 Citation Style

- Citation style: IEEE (adapted FAPS style)
- Reference sources in the text using numbers in square brackets: [xy]
- Multiple consecutive citations use separate brackets
- The bibliography is ordered chronologically and numbered consecutively
- Only actually cited works appear in the bibliography

0.4.2 Citation Placement

- A citation before the period refers to the specific sentence.
- A citation after the period of the last sentence refers to the entire paragraph.

0.4.3 Direct Quotes

- Must be letter-perfect and character-perfect reproductions
- Enclose in quotation marks; cite the source at the end

1 Introduction

1.1 Motivation and Context

- Tendon-driven robots: compliant design, reduced distal inertia, complex actuation modeling
- Reinforcement learning as a data-driven alternative to analytical control
- GPU-accelerated simulation (Isaac Sim + IsaacLab) as the enabling platform
- Frame the PA as the simulation foundation of a larger textile-sorting project at FAPS

1.2 Problem Statement and Objectives

- **Objective 1:** Validated tendon-driven robot model in Isaac Sim
- **Objective 2:** Progressive RL-based manipulation (reach then cube place), comparing PD vs tendon actuation
- **Objective 3:** Documented infrastructure foundation for the master thesis
- Scope exclusion: no cloth, no sorting, no sim-to-real, no wrist transfer (deferred to MA)

1.3 Thesis Outline

- Brief chapter summary for reader orientation

2 Background

2.1 Reinforcement Learning for Robotic Manipulation (2.1)

2.2 Physics Simulation in Isaac Sim 5.1 (2.2)

2.3 NVIDIA Isaac Sim and IsaacLab Architecture (2.3)

2.4 Tendon-Driven Robot Kinematics and Modeling (2.4)

3 State of the Art and Research Gap

3.1 GPU-Accelerated RL for Manipulation (3.1)

3.2 Tendon- and Cable-Driven Robot Simulation (3.2)

3.3 Workspace Analysis Methods (3.3)

3.4 Identified Research Gap (3.4)

3.5 Success Criteria (3.5)

4 Methodology

4.1 Overview of Approach (4.1)

4.2 Physical Robot Design (4.2)

4.2.1 Kinematic Chain and Degrees of Freedom (4.2.1)

4.2.2 Antiparallelogram Elbow Mechanism (4.2.2)

4.2.3 Cable-Driven Wrist (4.2.3)

4.2.4 Drive System and Tendon Routing (4.2.4)

4.3 Simulation Model Construction (4.3)

4.3.1 Mesh Processing Pipeline (4.3.1)

4.3.2 Articulation and Physics Configuration (4.3.2)

4.3.3 Disc Approximation Variant (4.3.3)

4.3.4 Physical Antiparallelogram Variant (4.3.4)

4.3.5 Five-DoF Assembly (4.3.5)

4.4 Tendon Actuation in Simulation (4.4)

4.4.1 Jacobian-Transpose Torque Mapping (4.4.1)

4.4.2 Physical Body-Force Tendon Model (4.4.2)

4.4.3 Actuation Mode Comparison (4.4.3)

4.4.4 Motor Specifications and Simulation Parameters (4.4.4)

4.5 Model Validation (4.5)

4.5.1 Step-Response Testing Protocol (4.5.1)

4.5.2 Monte Carlo Workspace Analysis (4.5.2)

4.6 Reinforcement Learning Task Formulation (4.6)

4.6.1 Simulation Environment (4.6.1)

4.6.2 Reach Task (4.6.2)

4.6.3 Cube Place Task (4.6.3)

4.6.4 Cube Sort Task — Outlook (4.6.4)

4.7 Training Infrastructure (4.7)

4.7.1 PPO Configuration (4.7.1)

4.7.2 Simulation Parameters (4.7.2)

4.7.3 Software and Hardware (4.7.3)

5 Results

5.1 Model Validation Results (5.1)

5.2 Reach Task Results (5.2)

5.3 Cube Place Results (5.3)

5.4 Cube Sort Preliminary (5.4)

5.5 Summary of Key Results (5.5)

6 Discussion

6.1 Evaluation of Outcomes (6.1)

6.2 Comparison with Related Work (6.2)

6.3 Limitations of the Study (6.3)

6.4 Practical Implications (6.4)

6.5 Recommendations and Lessons Learned (6.5)

7 Conclusion

7.1 Conclusion (7.1)

7.2 Outlook (7.2)

8 Appendices

- Appendix A: Simulation and Control Parameters
- Appendix B: PPO Hyperparameters
- Appendix C: Cube Place Reward Terms

Bibliography